

TECHNICAL DESIGN DOCUMENT (TDD)

System Forensic Scanner (SFS)

Version: 1.0

Status: Design Approved

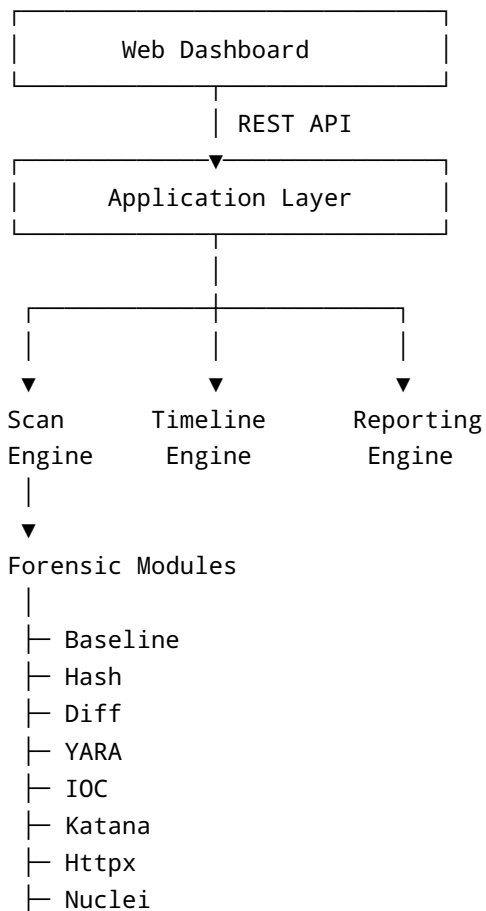
Language: Golang 1.25+

Database: PostgreSQL 17+

Architecture: Modular Monolith

Deployment: Docker / Native Binary

1. SYSTEM ARCHITECTURE



```
| Database
| Risk Engine
```

2. PROJECT STRUCTURE

```
sfs/
|
| cmd/
|   |
|   | api/
|   | worker/
|
| internal/
|   |
|   | api/
|   | auth/
|   | baseline/
|   | hash/
|   | diff/
|   | yara/
|   | ioc/
|   | katana/
|   | nuclei/
|   | httpx/
|   | forensicdb/
|   | timeline/
|   | risk/
|   | report/
|   | project/
|   | storage/
|   | worker/
|
| web/
|
| signatures/
|
| reports/
|
| temp/
|
| logs/
|
| configs/
```

3. CORE DEPENDENCIES

Backend

```
gin-gonic/gin  
gorm.io/gorm  
gorm.io/driver/postgres
```

YARA

```
github.com/hillu/go-yara/v4
```

Diff

```
github.com/pmezard/go-difflib
```

Filesystem

```
github.com/spf13/afero
```

PDF

```
github.com/jung-kurt/gofpdf
```

ZIP

```
archive/zip  
compress/gzip
```

4. DATABASE DESIGN

PostgreSQL

Schema:

```
CREATE SCHEMA sfs;
```

5. TABLE PROJECTS

```
CREATE TABLE projects (  
  id UUID PRIMARY KEY,  
  name VARCHAR(255),  
  target_type VARCHAR(100),  
  target_path TEXT,  
  baseline_path TEXT,  
  created_at TIMESTAMP,  
  updated_at TIMESTAMP  
);
```

6. TABLE SCANS

```
CREATE TABLE scans (  
  id UUID PRIMARY KEY,  
  project_id UUID,  
  status VARCHAR(50),  
  score INTEGER,  
  verdict VARCHAR(50),  
  started_at TIMESTAMP,  
  finished_at TIMESTAMP  
);
```

7. TABLE FINDINGS

```
CREATE TABLE findings (  
  id UUID PRIMARY KEY,  
  scan_id UUID,  
  module VARCHAR(100),  
  severity VARCHAR(50),  
  title TEXT,  
  description TEXT,  
  evidence JSONB  
);
```

8. TABLE EVIDENCE

```
CREATE TABLE evidence (  
  id UUID PRIMARY KEY,  
  scan_id UUID,  
  category VARCHAR(100),  
  path TEXT,  
  metadata JSONB,  
  created_at TIMESTAMP  
);
```

9. TABLE TIMELINE

```
CREATE TABLE timeline (  
  id UUID PRIMARY KEY,  
  scan_id UUID,  
  event_time TIMESTAMP,  
  source VARCHAR(100),  
  title TEXT,  
  details JSONB  
);
```

10. TABLE REPORTS

```
CREATE TABLE reports (  
  id UUID PRIMARY KEY,  
  scan_id UUID,  
  json_path TEXT,  
  html_path TEXT,  
  pdf_path TEXT,  
  generated_at TIMESTAMP  
);
```

11. SCAN PIPELINE

```
Start Scan  
↓  
Baseline Compare  
↓
```

```
Hash Scan
↓
Diff Scan
↓
YARA Scan
↓
IOC Scan
↓
Database Forensic
↓
Katana
↓
Httpx
↓
Nuclei
↓
Timeline Build
↓
Risk Calculation
↓
Generate Report
↓
Complete
```

12. BASELINE ENGINE DESIGN

Input:

```
Baseline Folder
Target Folder
```

Output:

```
{
  "added": [],
  "deleted": [],
  "modified": []
}
```

Hash:

```
SHA256
```

Worker:

20 Concurrent Workers

13. HASH ENGINE

Output:

```
{
  "path": "plugins/test.php",
  "hash": "...",
  "size": 1024,
  "mtime": "..."
}
```

Storage:

evidence.metadata

14. DIFF ENGINE

Trigger:

Modified File

Store:

```
{
  "before": "clean",
  "after": "modified",
  "diff": "unified diff"
}
```

Limit:

Max 1 MB per diff

15. YARA ENGINE

Rule Sources

```
signatures/yara/
```

Cache:

```
24 Hours
```

Workflow:

```
Load Rules
↓
Compile Rules
↓
Scan Files
↓
Store Matches
```

Output:

```
{
  "rule": "PHP_Webshell",
  "severity": "critical",
  "file": "shell.php"
}
```

16. IOC ENGINE

Regex Database

```
eval(
  base64_decode(
    shell_exec(
      system(
        passthru(
          assert(
            preg_replace
            gzinflate(
```

Scoring:

1 IOC = 10 points

17. DATABASE FORENSIC ENGINE

Supported:

```
.sql  
.sql.gz  
.zip
```

Process:

```
Import Dump  
↓  
Parse SQL  
↓  
Extract Tables  
↓  
Analyze
```

Detection:

```
Suspicious User  
Privilege Escalation  
New Admin  
Unknown Submission
```

18. TIMELINE ENGINE

Sources:

Filesystem

Apache Logs

Application Logs

Database Events

Normalize:

```
{
  "timestamp": "",
  "source": "",
  "event": "",
  "details": {}
}
```

Sort:

Ascending Timestamp

19. KATANA ENGINE

Execution:

```
katana
-jsonl
-depth 10
```

Modes:

Anonymous

Cookie Based

Authenticated

Output:

```
{
  "url": "",
  "type": "endpoint"
}
```

Store:

evidence

20. HTTPX ENGINE

Execution:

```
httpx  
-json  
-tech-detect
```

Extract:

```
Server  
Framework  
TLS  
Headers  
WAF  
Cloudflare
```

21. NUCLEI ENGINE

Execution:

```
nuclei  
-json  
-severity critical,high,medium
```

Template Location:

```
signatures/nuclei/
```

Output:

```
{  
  "template": "",  
  "severity": "",  
  "matched": ""  
}
```

22. RISK ENGINE

Weights

```
Webshell = 100
YARA = 90
Core Modified = 80
Unknown Plugin = 70
Critical CVE = 60
IOC = 40
```

Formula:

```
Risk Score =
Total Findings Weight /
Maximum Weight
```

Verdict:

```
SAFE

SUSPICIOUS

COMPROMISED

CRITICALLY COMPROMISED
```

23. REPORT ENGINE

Generate:

```
report.json
report.html
report.pdf
```

Storage:

```
reports/{scan-id}/
```

24. API DESIGN

Base URL

/api/v1

Projects

GET /projects

POST /projects

GET /projects/:id

DELETE /projects/:id

Scans

POST /scans/start

GET /scans/:id

GET /scans/:id/report

Evidence

GET /evidence

GET /evidence/:id

Timeline

GET /timeline/:scanId

25. WORKER DESIGN

Worker Pools

```
Hash Worker = 20
```

```
YARA Worker = 10
```

```
IOC Worker = 20
```

```
Diff Worker = 5
```

Queue:

```
chan ScanTask
```

26. SECURITY DESIGN

Read Only

No Delete

No Rename

No Patch

No Execute Remote Code

No Reverse Shell

No Exploit Module

27. REPORT DIRECTORY

```
reports/
```

```
scan-id/
```

```
├─ report.json
```

```
├─ report.html
```

```
├─ report.pdf
```

```
└─ findings.json
```

```
|─ timeline.json
|─ evidence/
```

28. FUTURE PLUGIN DESIGN

```
plugins/

ojs/

wordpress/

laravel/

moodle/

generic/
```

Interface:

```
type ScannerPlugin interface {
    Detect()
    Analyze()
    Report()
}
```

29. DEPLOYMENT

Docker

```
docker compose up -d
```

Native

```
./sfs-api
```

30. MVP ACCEPTANCE

✓ Scan 100.000 file

- ✓ Compare baseline
- ✓ Parse SQL Dump
- ✓ YARA Detection
- ✓ IOC Detection
- ✓ Katana Integration
- ✓ Httpx Integration
- ✓ Nuclei Integration
- ✓ Timeline Reconstruction
- ✓ Risk Scoring
- ✓ JSON Report
- ✓ HTML Report
- ✓ PDF Report
- ✓ Windows Support
- ✓ Linux Support